

# Machine Learning — Exam Q&A Study Guide

25 Questions covering supervised learning, classification, KNN, SVM, PCA, logistic regression, gradient descent, and ensemble learning — with full worked answers and explanations.

## Section 1: Machine Learning — General

Question 1 | Single Choice | **Easy** | Time: 2 Minutes

**Q1. Which of the following best describes supervised learning?**

- ☐ A. The model learns patterns from unlabeled data without any guidance
- ☒ **B. The model learns a mapping from inputs to outputs using labeled training data**
- ☐ C. The model learns by interacting with an environment and receiving rewards or penalties
- ☐ D. The model clusters data points into groups based on similarity

✓ **Correct Answer: B**

**A — Wrong:** This describes unsupervised learning.

**B — Correct:** Supervised learning uses labeled input-output pairs to train the model.

**C — Wrong:** This describes reinforcement learning.

**D — Wrong:** This describes unsupervised clustering.

## Section 2: Classification

Question 2 | Single Choice | **Easy** | Time: 2 Minutes

**Q2. Which of the following best describes a classification problem in machine learning?**

- ☐ A. Predicting a continuous numerical output for a given input
- ☐ B. Grouping similar data points together without using any labels
- ☒ **C. Predicting which discrete category or class a new input belongs to**
- ☐ D. Reducing the number of features in the dataset

✓ **Correct Answer: C**

**A — Wrong:** This describes a regression problem.

**B — Wrong:** This describes unsupervised clustering.

**C — Correct:** Classification assigns an input to one of a fixed set of discrete classes.

**D — Wrong:** This describes dimensionality reduction such as PCA.

## Section 3: Classification Metrics

**Q3. In binary classification, which of the following correctly defines Precision?**

- ☐ A.  $TP / (TP + FN)$
- ☒ B.  $TP / (TP + FP)$
- ☐ C.  $(TP + TN) / (TP + TN + FP + FN)$
- ☐ D.  $TN / (TN + FP)$

✓ **Correct Answer: B**

| Formula                    | Metric Name          |
|----------------------------|----------------------|
| $TP / (TP + FP)$           | Precision            |
| $TP / (TP + FN)$           | Recall (Sensitivity) |
| $(TP + TN) / \text{Total}$ | Accuracy             |
| $TN / (TN + FP)$           | Specificity          |

- A — Wrong:** This is the formula for Recall.
- B — Correct:** Precision measures how many predicted positives were actually positive.
- C — Wrong:** This is the formula for Accuracy.
- D — Wrong:** This is the formula for Specificity.

## Section 4: K-Nearest Neighbors (KNN)

**Q4. Which of the following statements correctly describe KNN? Select all that apply.**

- ☐ A. KNN performs most of its computational work during training to learn an explicit decision function.
- ☒ B. KNN is considered a Lazy Learning algorithm because it postpones most of the computation until a prediction is requested.
- ☒ C. In KNN, the training phase mainly involves storing the training instances rather than learning a fixed parametric model.
- ☐ D. Increasing the training dataset always makes KNN prediction faster because more neighbors are available.

✓ **Correct Answers: B, C**

- A — Wrong:** KNN does almost no computational work during training.
- B — Correct:** KNN delays all computation until prediction time, which is the definition of Lazy Learning.
- C — Correct:** The training phase in KNN is simply storing the data — no model is explicitly learned.
- D — Wrong:** More training data means more distance calculations at prediction time, making it slower not faster.

**Q5. Suppose you are using KNN for a binary classification problem. You observe that with  $K = 1$ , the model perfectly classifies the training data but performs poorly on unseen data. Which changes are likely to help? *Select all that apply.***

- ☒ A. Increase K to make the prediction less sensitive to individual noisy training points.
- ☐ B. Decrease K further because a smaller K always improves generalization.
- ☒ C. Use an appropriate feature-scaling method before calculating distances.
- ☐ D. Increase K indefinitely until the model achieves 100% training accuracy.

✓ **Correct Answers: A, C**

- A — Correct:** Higher K produces a smoother decision boundary and reduces overfitting.
- B — Wrong:** Smaller K increases sensitivity to noise, making overfitting worse.
- C — Correct:** Feature scaling ensures all features contribute fairly to the distance calculation.
- D — Wrong:** Very large K leads to underfitting and does not produce 100% training accuracy.

## Section 5: Support Vector Machines (SVM)

**Q6. In a Soft-Margin SVM, the optimization problem is shown below. Suppose we increase C significantly while keeping the dataset and kernel fixed. Which of the following is the most accurate consequence?**

```
minimize:
(1/2)||w||^2 + C * sum(xi_i)

subject to:
y_i(w^T x_i + b) >= 1 - xi_i
xi_i >= 0
```

- ☐ A. The SVM will prioritize maximizing the margin and tolerate more margin violations.
- ☒ B. The SVM will penalize margin violations more heavily and tend to fit the training data more strictly.
- ☐ C. The SVM will automatically change the kernel from linear to nonlinear.
- ☐ D. The SVM will make the value of  $xi_i$  larger for every training point.

✓ **Correct Answer: B**

- A — Wrong:** Large C means less tolerance for violations, not more.
- B — Correct:** Large C increases the penalty per violation, forcing a stricter fit on the training data.
- C — Wrong:** C has no effect on the kernel type.
- D — Wrong:** Large C forces  $xi_i$  to be smaller, not larger.

**Q7. Consider a training example for which  $y_i(w^T x_i + b) = 0.6$ . Using the Soft-Margin SVM constraint  $y_i(w^T x_i + b) \geq 1 - \xi_i$ , assuming the smallest possible value of  $\xi_i$  is used, which statement is correct?**

- ☐ A.  $\xi_i = 0$ , and the point is outside the margin.
- ☒ B.  $\xi_i = 0.4$ , and the point is correctly classified but lies inside the margin.
- ☐ C.  $\xi_i = 1.6$ , and the point is misclassified.
- ☐ D.  $\xi_i = 0.6$ , and the point lies exactly on the margin.

✓ **Correct Answer: B**

Calculation:

```
0.6 >= 1 - xi_i
xi_i >= 1 - 0.6
xi_i = 0.4 (minimum value)
```

Since  $0 < \xi_i < 1$ , the point is correctly classified but lies inside the margin.

**Q8. Which of the following statements correctly describe the relationship between C and the Soft-Margin SVM? Select all that apply.**

- ☒ A. A larger C makes violations of the margin more expensive.
- ☒ B. A smaller C generally corresponds to stronger regularization and allows the model to tolerate more violations.
- ☒ C. As C becomes very large, the Soft-Margin SVM can approach Hard-Margin behavior when the data is linearly separable.
- ☐ D. Increasing C always makes a Linear SVM decision boundary nonlinear.
- ☐ E. C controls the distance between the two margin hyperplanes directly.

✓ **Correct Answers: A, B, C**

- A — Correct:** Larger C assigns a higher penalty per margin violation.
  - B — Correct:** Smaller C allows more violations, which acts as stronger regularization.
  - C — Correct:** As C approaches infinity, no violations are allowed, which is equivalent to Hard-Margin SVM.
  - D — Wrong:** C does not change the kernel type or the linearity of the boundary.
  - E — Wrong:** C controls the violation penalty, not the margin width directly.
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**Q9. In the dual formulation of the Soft-Margin SVM, the Lagrange multipliers satisfy  $0 \leq \alpha_i \leq C$ . Which statements correctly follow from this relationship? Select all that apply.**

- ☒ **A.**  $C$  acts as an upper bound on each  $\alpha_i$ .
- ☒ **B.** A training point with  $\alpha_i = 0$  does not contribute directly to the SVM decision function.
- ☒ **C.** A point with  $0 < \alpha_i < C$  is typically located exactly on the margin.
- ☐ **D.** A point with  $\alpha_i = C$  must always be correctly classified and lie outside the margin.
- ☒ **E.** The constraint  $\alpha_i \leq C$  comes from the Slack Variable term  $C * \sum(x_i)$  in the primal optimization problem.

✓ **Correct Answers: A, B, C, E**

**A — Correct:** By definition of the dual constraint.

**B — Correct:** Non-support vectors have  $\alpha_i = 0$  and do not influence the decision boundary.

**C — Correct:** Points exactly on the margin are support vectors with  $0 < \alpha_i < C$ .

**D — Wrong:**  $\alpha_i = C$  indicates a margin violator, which may be misclassified.

**E — Correct:** The upper bound  $C$  originates from the slack penalty term in the primal problem.

## Section 6: Principal Component Analysis (PCA)

**Q10. What is the main purpose of Principal Component Analysis (PCA)?**

- ☐ **A.** To increase the number of features in a dataset
- ☒ **B.** To reduce the dimensionality of data while preserving as much variance as possible
- ☐ **C.** To convert a classification problem into a regression problem
- ☐ **D.** To eliminate the need for training data

✓ **Correct Answer: B**

PCA identifies directions of maximum variance and projects the data onto a lower-dimensional space while retaining as much information as possible.

**Q11. Which statement correctly describes the relationship between PCA, eigenvectors, and eigenvalues?**

- ☐ **A.** Eigenvalues determine the direction of the new axes, while eigenvectors determine the variance
- ☒ **B.** Eigenvectors determine the directions of the new axes, while eigenvalues indicate the amount of variance captured in those directions
- ☐ **C.** Both eigenvectors and eigenvalues represent the original features of the dataset
- ☐ **D.** Eigenvectors are used to calculate the target variable, while eigenvalues are used for normalization

✓ **Correct Answer: B**

Eigenvectors define the direction of each principal component axis. Eigenvalues measure the amount of variance explained by each corresponding principal component.

**Q12. Suppose a dataset has four principal components with eigenvalues  $\lambda_1 = 8$ ,  $\lambda_2 = 4$ ,  $\lambda_3 = 2$ ,  $\lambda_4 = 1$ . If we want to retain at least 85% of the total variance, how many principal components should we keep?**

- ☐ A. 1
- ☐ B. 2
- ☒ C. 3
- ☐ D. 4

✓ **Correct Answer: C**

Calculation:

Total variance =  $8 + 4 + 2 + 1 = 15$

PC1:  $8 / 15 = 53.3\%$

PC1 + PC2:  $12 / 15 = 80.0\%$  -> below 85%

PC1 + PC2 + PC3:  $14 / 15 = 93.3\%$  -> above 85% (keep 3)

## Section 7: Logistic Regression

**Q13. In Logistic Regression, the model computes  $z = w^T x + b$  and  $p = 1 / (1 + e^{-z})$ . If the model predicts  $p = 0.9$  for a sample whose true label is  $y = 0$ , which statement is MOST accurate?**

- ☐ A. The prediction is correct because  $p > 0.5$ .
- ☐ B. The model will have a small Binary Cross-Entropy loss because  $p$  is close to 1.
- ☒ C. The model will have a large Binary Cross-Entropy loss because it is highly confident in an incorrect class.
- ☐ D. The model cannot produce  $p = 0.9$  when  $y = 0$ .

✓ **Correct Answer: C**

Calculation:

```
Loss = -[ y * log(p) + (1 - y) * log(1 - p) ]  
= -[ 0 * log(0.9) + 1 * log(0.1) ]  
= -log(0.1)  
≈ 2.30 (very large loss)
```

The model is 90% confident in the wrong class, which results in a very large penalty.

Question 14 | Single Choice | **Hard** | Time: 5 Minutes

**Q14. For a single training example in Logistic Regression, suppose  $y = 1$ ,  $p = 0.8$ ,  $x = [2, -3]^T$ . Given that  $dL/dw = (p - y) * x$ , what is the gradient with respect to  $w$ ?**

- ☐ A.  $[0.4, -0.6]^T$
- ☒ B.  $[-0.4, 0.6]^T$
- ☐ C.  $[-0.2, 0.3]^T$
- ☐ D.  $[0.2, -0.3]^T$

✓ **Correct Answer: B**

Calculation:

$$(p - y) = 0.8 - 1.0 = -0.2$$

$$\begin{aligned} dL/dw &= -0.2 * [2, -3]^T \\ &= [-0.4, 0.6]^T \end{aligned}$$

## Section 8: Gradient Descent

Question 15 | Single Choice | **Easy** | Time: 2 Minutes

**Q15. What is the main goal of Gradient Descent?**

- ☐ A. Increase the number of features
- ☒ B. Minimize the loss/cost function
- ☐ C. Maximize the learning rate
- ☐ D. Increase the training dataset

✓ **Correct Answer: B**

Gradient Descent is an optimization algorithm that minimizes the cost function by iteratively updating model parameters in the direction of the negative gradient.

Question 16 | Single Choice | **Easy** | Time: 2 Minutes

**Q16. In Gradient Descent, what does the learning rate alpha control?**

- ☐ A. The number of training examples
- ☐ B. The direction of the gradient
- ☒ C. The size of the parameter update step
- ☐ D. The value of the loss function

✓ **Correct Answer: C**

The learning rate alpha determines how large each parameter update step is during training.

**Q17. Which of the following are true about Gradient Descent? *Select all that apply.***

- ☒ A. It is an optimization algorithm.
- ☒ B. It can be used to minimize a cost function.
- ☐ C. It always finds the global minimum.
- ☒ D. It updates model parameters iteratively.

✓ **Correct Answers: A, B, D**

**A — Correct:** Gradient Descent is fundamentally an optimization algorithm.

**B — Correct:** Its primary purpose is to minimize the cost function.

**C — Wrong:** Non-convex cost functions contain local minima and saddle points that can prevent GD from reaching the global minimum.

**D — Correct:** Parameters are updated step by step each iteration.

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**Q18. Consider the update rule  $\theta := \theta - \alpha * (dJ(\theta) / d_{\theta})$ . If the gradient is positive, what happens to  $\theta$ ?**

- ☐ A.  $\theta$  increases
- ☒ B.  $\theta$  decreases
- ☐ C.  $\theta$  remains unchanged
- ☐ D.  $\theta$  becomes zero

✓ **Correct Answer: B**

Calculation:

```
theta_new = theta - alpha * (positive value)
= theta - (something positive)
```

Therefore  $\theta$  decreases and moves in the negative direction toward the minimum.

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**Q19. Which situations can occur if the learning rate is too large? *Select all that apply.***

- ☒ A. The algorithm may overshoot the minimum.
- ☒ B. The cost function may oscillate.
- ☒ C. The algorithm may diverge.
- ☐ D. The algorithm will always converge faster.

✓ **Correct Answers: A, B, C**

**A — Correct:** Large steps can cause the update to jump past the minimum.

**B — Correct:** Repeated overshooting causes the loss to oscillate up and down.

**C — Correct:** Extreme overshooting can cause the loss to grow indefinitely (diverge).

**D — Wrong:** A large learning rate does not guarantee faster convergence and often prevents it entirely.

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**Q20. Suppose Gradient Descent reaches a point where  $dJ / d_{\theta} = 0$ . What does this indicate?**

- ☐ A. The parameter must be at the global minimum.
- ☒ B. There is no gradient-based change at that point.
- ☐ C. The learning rate is zero.
- ☐ D. The cost function is necessarily constant everywhere.

✓ **Correct Answer: B**

Calculation:

```
theta := theta - alpha * 0 = theta
```

No change occurs. This could correspond to a global minimum, a local minimum, or a saddle point — not necessarily the global minimum.

**Q21. Which statements correctly describe Batch Gradient Descent? *Select all that apply.***

- ☒ A. It computes the gradient using the entire training dataset for each update.
- ☒ B. Each update can be computationally expensive for very large datasets.
- ☒ C. The gradient estimate is generally less noisy than in SGD.
- ☐ D. It updates the parameters after processing exactly one training example.

✓ **Correct Answers: A, B, C**

**A — Correct:** The full dataset is used per update — that is the definition of Batch GD.

**B — Correct:** Processing all data per step is expensive for large datasets.

**C — Correct:** Using all data produces a stable and accurate gradient estimate.

**D — Wrong:** Processing exactly one example per update describes Stochastic Gradient Descent (SGD).

**Q22. Consider a one-dimensional cost function where  $\theta = 4$ ,  $dJ / d_{\theta} = 6$ ,  $\alpha = 0.1$ . Which statements are correct? *Select all that apply.***

- ☒ A. The new parameter is 3.4.
- ☒ B. The parameter moves in the negative direction.
- ☒ C. The update is -0.6.
- ☐ D. The new parameter is 4.6.

✓ **Correct Answers: A, B, C**

Calculation:

```
theta_new = theta - alpha * (dJ / d_theta)
= 4 - (0.1)(6)
= 4 - 0.6
= 3.4
```

Update = -0.6 (correct). Direction is negative since  $\theta$  decreased from 4 to 3.4 (correct). D is wrong: 4.6 would result from adding the gradient instead of subtracting it.

**Q23. A model is being trained using Gradient Descent. During training you observe the loss sequence  $J(\theta)$ : 10.2 -> 8.7 -> 7.1 -> 9.4 -> 6.8 -> 11.3. Which explanations are plausible? Select all that apply.**

- ☒ **A.** The learning rate may be too large.
- ☒ **B.** The optimization process may be overshooting the minimum.
- ☐ **C.** The cost function is necessarily non-convex.
- ☒ **D.** Reducing the learning rate may help stabilize training.
- ☐ **E.** The model must be suffering from underfitting.

✓ **Correct Answers: A, B, D**

- A — Correct:** Erratic jumps in the loss are a classic sign of a learning rate that is too large.
- B — Correct:** The loss decreasing then jumping back up indicates the optimizer is overshooting the minimum.
- C — Wrong:** This behavior can occur even on convex functions when the learning rate is too large.
- D — Correct:** Reducing alpha is the standard remedy for this type of instability.
- E — Wrong:** Underfitting refers to model capacity, not to this loss oscillation pattern.

## Section 9: Ensemble Learning

**Q24. Which of the following best describes Ensemble Learning?**

- ☐ **A.** Training a single complex model on the entire dataset to maximize accuracy
- ☒ **B.** Combining multiple models to produce a stronger overall prediction than any individual model
- ☐ **C.** Reducing the number of features before training a single model
- ☐ **D.** Using one model to label unlabeled data for use by a second model

✓ **Correct Answer: B**

Ensemble Learning combines the predictions of multiple base models to reduce variance, reduce bias, or improve overall prediction accuracy compared to using any single model alone.

**Q25. Which of the following statements correctly describe Random Forest? Select all that apply.**

- ☒ **A.** It is an ensemble method that builds multiple decision trees.
- ☒ **B.** Each tree is trained on a random subset of the training data using bootstrapping (sampling with replacement).
- ☐ **C.** It uses boosting to sequentially correct the errors of previous trees.
- ☒ **D.** The final classification prediction is determined by majority voting among all trees.
- ☒ **E.** At each split, a random subset of features is considered to increase diversity among trees.

✓ **Correct Answers: A, B, D, E**

**A — Correct:** Random Forest is an ensemble of decision trees.

**B — Correct:** Bootstrapping creates a different training subset for each tree, increasing diversity.

**C — Wrong:** Random Forest uses bagging (parallel training), not boosting. Boosting is used in AdaBoost and Gradient Boosting.

**D — Correct:** For classification, the final output is the majority vote across all trees.

**E — Correct:** Random feature selection at each node is a core mechanism that reduces correlation between trees.

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## Exam Summary Table

| #  | Topic                                           | Type            | Difficulty | Time  |
|----|-------------------------------------------------|-----------------|------------|-------|
| 1  | ML General — Supervised Learning                | Single Choice   | Easy       | 2 min |
| 2  | Classification — Definition                     | Single Choice   | Easy       | 2 min |
| 3  | Classification Metrics — Precision              | Single Choice   | Easy       | 2 min |
| 4  | KNN — Lazy Learning                             | Multiple Choice | Medium     | 5 min |
| 5  | KNN — Overfitting (K=1)                         | Multiple Choice | Medium     | 4 min |
| 6  | SVM — Effect of large C                         | Single Choice   | Medium     | 4 min |
| 7  | SVM — Slack variable $\xi_i$                    | Single Choice   | Medium     | 4 min |
| 8  | SVM — C and regularization                      | Multiple Choice | Medium     | 5 min |
| 9  | SVM — Dual formulation $\alpha_i$               | Multiple Choice | Hard       | 6 min |
| 10 | PCA — Main purpose                              | Single Choice   | Easy       | 2 min |
| 11 | PCA — Eigenvectors and Eigenvalues              | Single Choice   | Easy       | 3 min |
| 12 | PCA — Variance retention (calculation)          | Single Choice   | Medium     | 4 min |
| 13 | Logistic Regression — Loss intuition            | Single Choice   | Medium     | 3 min |
| 14 | Logistic Regression — Gradient calculation      | Single Choice   | Hard       | 5 min |
| 15 | Gradient Descent — Main goal                    | Single Choice   | Easy       | 2 min |
| 16 | Gradient Descent — Learning rate role           | Single Choice   | Easy       | 2 min |
| 17 | Gradient Descent — True statements              | Multiple Choice | Easy       | 3 min |
| 18 | Gradient Descent — Positive gradient effect     | Single Choice   | Medium     | 3 min |
| 19 | Gradient Descent — Large learning rate          | Multiple Choice | Medium     | 4 min |
| 20 | Gradient Descent — Zero gradient                | Single Choice   | Medium     | 3 min |
| 21 | Gradient Descent — Batch GD                     | Multiple Choice | Medium     | 4 min |
| 22 | Gradient Descent — Parameter update calculation | Multiple Choice | Hard       | 5 min |
| 23 | Gradient Descent — Erratic loss behavior        | Multiple Choice | Very Hard  | 6 min |
| 24 | Ensemble Learning — Definition                  | Single Choice   | Easy       | 3 min |
| 25 | Ensemble Learning — Random Forest               | Multiple Choice | Medium     | 4 min |

**Total Estimated Time: ~94 minutes**